

Our Green World: Resources and Conservation

Complete Answer Key — Science Lesson 11

This reference document provides comprehensive, verified answers to all activities, fill-in-the-blanks, in-class questions, end-of-chapter exercises, and Higher Order Thinking Skills (HOTS) questions in **Lesson 11: Our Green World** from the Class 6 Science textbook. All answers are grounded strictly in the academic content, principles of conservation, and waste management concepts presented across the chapter's pages.

Recap Time — Keeping Air & Water Clean (Page 138)

Draw a smiley (■) for the actions that will help keep air and water clean.

1. We must grow plants.

■ (Smiley) — Plants purify the air by absorbing carbon dioxide and releasing oxygen.

2. We must wash clothes in rivers.

■ (No Smiley) — Soap and detergents pollute river water, making it toxic for aquatic life.

3. We must not throw waste in river, pond, lake or sea.

■ (Smiley) — Preventing waste disposal in waterbodies keeps aquatic habitats clean and safe.

4. We must not burn garbage.

■ (Smiley) — Avoiding burning garbage prevents harmful smoke and toxic gases from polluting the air.

5. We must cut down trees.

■ (No Smiley) — Cutting down trees reduces oxygen levels, destroys habitats, and causes soil erosion.

6. We must not use a bicycle to go to nearby places.

■ (No Smiley) — We should use a bicycle for short trips to save fuel and reduce air pollution.

7. We must use public transport as often as we can.

■ **(Smiley)** — Using public transport reduces the number of vehicles on the road, lowering overall smoke emissions.

Let's Start — Obtained From Nature (Page 139)

A list of things is given below. Underline the things that you think are obtained from nature.

List of things: sunlight, rocks, air, fruits, flowers, wood, glass, honey, paper, rubber, water, milk, plastic, cement, medicine

Underlined items (Obtained from Nature):

- **Sunlight** (Direct natural energy from the sun)
- **Rocks** (Natural geological formations)
- **Air** (Gaseous envelope of the Earth)
- **Fruits & Flowers** (Direct botanical products)
- **Wood** (Obtained directly from natural tree trunks)
- **Honey** (Produced naturally by honeybees)
- **Water** (Natural liquid hydrosphere resource)
- **Milk** (Secreted naturally by mammals)

Note on other items: Glass, paper, rubber, plastic, cement, and medicine are human-made, synthetic, or highly processed factory-manufactured items, and therefore are not considered raw natural things directly obtained from nature.

My Eureka Moment — Plastic & Aquatic Life (Page 142)

Read the lines from the song 'The Plastic War' by Grammy winner Ricky Kej, and answer the questions.

1. Why should we stop throwing plastic bags away?

We must stop throwing plastic bags away because plastic is non-biodegradable (it does not decay or go under). Discarded plastic bags float on the surface of waterbodies or sink to the bottom, choking rivers, lakes, and oceans, and persisting in the environment for hundreds of years as long-lasting pollution.

2. Does it affect animals and plants living in water? Find out how.

Yes, plastic bags severely affect aquatic animals and plants. Aquatic creatures like sea turtles, fish, and blue whales often mistake floating plastic bags for food (such as jellyfish) and swallow them. This causes choking, stomach blockages, and starvation. Additionally, plastic accumulation on water beds blocks vital sunlight and oxygen from reaching aquatic plants, severely hindering their growth and survival.

3. Make an attractive design of a bag to be made by using colourful pieces of old clothes.

Creative Activity Solution (Eco-Friendly Cloth Bag):

To make an upcycled cloth bag for shopping, follow these steps:

- Material:** Gather clean, colorful remnants of old cotton t-shirts, kurtas, or denim trousers.
- Cutting:** Cut two identical rectangular sheets (e.g., 40 cm x 35 cm) to serve as the front and back panels.
- Sewing:** Place the panels together and sew along the left, right, and bottom edges with a strong double stitch.
- Hemming:** Fold the top opening edge outward by 2 cm and hem it down to form a clean, strong border.
- Handles:** Cut two strong cloth strips (50 cm x 6 cm), fold them lengthwise, sew them into flat straps, and stitch them securely to the top rims.
- Decoration:** Stitch colorful cloth patches or sew simple patterns with vibrant thread to make it highly attractive.

In Class — Fill in the Blanks (Page 144)

Question Sentence	Answer Fill-in
1. Soil can be protected by growing	plants (Vegetation roots hold soil firmly together)
2. Soil is a natural resource.	renewable (It forms continuously over very long periods but is renewable)
3. were one of the first things made by human beings.	Clay pots (Molded from natural mud and clay)

Exercises — Answer Key (Page 148)

I. Correct the following sentences.

1. Plastic is a natural resource.

Correction: Plastic is a **man-made** (or synthetic) material. (Or: *Soil/Air/Water is a natural resource.*)

2. Petrol is a renewable resource.

Correction: Petrol is a **non-renewable** resource. (It takes millions of years to form and cannot be renewed in a short period.)

3. Polar bears are found in the Gir forest of Gujarat.

Correction: **Asiatic lions** are found in the Gir forest of Gujarat. (Polar bears are native to the cold Arctic polar regions.)

4. Penguins were once found in many parts of India.

Correction: **Asiatic cheetahs** (or Tigers) were once found in many parts of India. (Penguins are flightless birds found in the Southern Hemisphere and Antarctica, never native to India.)

5. Discarded metal tumbler cannot be recycled.

Correction: A discarded metal tumbler **can** be recycled. (Metals are valuable materials that can be melted down and repurposed.)

II. Answer the following questions in one sentence.

1. Define natural resources.

Natural things that are useful to us are called **natural resources** (for example: plants, forests, rivers, sunlight, and air).

2. What do you mean by 'renewable natural resources'?

Renewable natural resources are those resources that are being renewed continuously by natural processes in a short period of time (such as air, water, and forests).

3. What is a non-renewable natural resource?

A **non-renewable natural resource** is a resource that is available in a limited amount and takes a very long period of time to form, meaning it cannot be renewed in a short span (such as coal, petrol, and diesel).

4. Why is it important to separate waste?

It is important to separate waste so that recyclable materials (like metal, paper, and plastic) can be easily processed in separate bags, while organic kitchen waste is kept aside to make useful agricultural compost.

5. List the animals that are struggling to survive in the Himalayan region.

Animals that are struggling to survive (endangered) in the Himalayan region include the **snow leopard**, **red panda**, **musk deer**, and the **Himalayan brown bear**.

III. Answer the following questions in detail.

1. List the steps that can be taken to prevent air pollution.

To effectively reduce and prevent air pollution, the following steps can be taken:

- **Active Commuting:** Walk or ride a bicycle for short distances to save fuel and eliminate emissions.
- **Vehicle Maintenance:** Ensure regular servicing of family cars, scooters, and motor vehicles to reduce toxic exhaust smoke.
- **Use Public Transport:** Travel by public buses, metro, or carpools as much as possible to limit the total number of vehicles on roads.
- **Industrial Regulation:** Do not permit smoke- and noise-producing factories or open-air garbage dumping near residential zones.
- **Public Awareness:** Educate community members on the dangers of burning household garbage and plastics.

2. What can we do to conserve water?

Water conservation can be achieved through daily habits and infrastructure modifications:

- **Fix Plumbing Leaks:** Always close taps tightly when not in use and immediately repair leaky pipes and taps.
- **Rainwater Harvesting:** Build small check dams or household collection tanks to catch and store clean rainwater during monsoons.
- **Reduce Chemical Use:** Minimize the use of synthetic laundry detergents and bath soaps, which pollute municipal wastewater.
- **Protect Waterbodies:** Prevent direct dumping of household garbage, plastic waste, or sewage into rivers, lakes, and oceans.

3. Write the steps that can be taken to reduce water pollution.

Reducing water pollution requires strict trash disposal discipline and civic action:

- **Proper Waste Separation:** Collect plastic, glass, and metal waste in designated bags and recycle them, ensuring they never reach waterways.
- **Avoid Waterway Washing:** Do not wash clothes, scrub household utensils, or bathe domestic animals in rivers, streams, or lakes.
- **Sewage Treatment:** Treat domestic sewage and toxic industrial effluents in treatment plants before discharging them into natural rivers.
- **Eco-friendly Farming:** Reduce chemical fertilizer and pesticide usage in crops, transitioning to compost to avoid toxic farm runoff into aquifers.

4. Polluted water can harm all kinds of living things. Justify this statement.

Water is essential for all life; therefore, water pollution causes widespread damage across ecosystems:

- **For Human Health:** Drinking or cooking with polluted water transmits deadly waterborne pathogenetic diseases, such as cholera, typhoid, dysentery, hepatitis A, and chronic diarrhoea.
- **For Aquatic Organisms:** Floating plastic debris chokes aquatic animals (turtles, fish, and blue whales), while sewage and chemicals block sunlight, consume dissolved oxygen, and cause mass suffocation of aquatic life.
- **For Soil and Crops:** Polluted irrigation water makes agricultural fields toxic, kills beneficial soil earthworms/microorganisms, and ruins crop quality.

5. How do plants prevent soil from being damaged by rainwater?

Plants protect topsoil from water-induced damage (soil erosion) through two primary biological mechanisms:

- **Root Anchoring:** The extensive root networks of trees and plants bind soil particles tightly together, acting as a physical net that prevents rainwater from easily washing the soil away.
- **Canopy Interception:** The foliage (leaves, canopy, and branches) of plants absorbs the direct force of falling raindrops, slowing down their speed. This cushions the impact on the bare ground, preventing the rain from dislodging and muddying the fertile topsoil.

IV. Draw and colour a forest. Why should forests be protected?

Drawing Guidelines:

1. Draw several tall trees with wide leafy crowns (using different shades of green) and strong wooden trunks (brown).
2. Draw bushes, shrubs, and wildflowers along the forest floor.
3. Add small animals such as a rabbit, deer, and birds perched on tree branches or flying.
4. Sketch a clean, sparkling blue stream winding through the forest floor.

Why Forests Must Be Protected:

- **Habitats:** Forests serve as natural homes (habitats) for millions of species of plants, birds, insects, and animals.
- **Oxygen Producers:** They act as the Earth's green lungs, releasing fresh oxygen and absorbing harmful carbon dioxide.
- **Soil & Water Balance:** Tree roots prevent soil erosion, while forest canopies contribute heavily to rain formation and keep the water cycle flowing.
- **Resource Providers:** They supply humans with valuable natural resources such as wood, timber, honey, seeds, and lifesaving medicines.

Think and Answer — Higher Order Thinking Skills (Page 149)

Read the passage carefully:

"Freshwater, although is renewed by rain, is in limited quantities and can be exhausted, at least temporarily. Freshwater in rivers, lakes, ponds and wells can dry if overused or due to very less rainfall. If we do not use water carefully, water can become scarce, at least till rain renews it."

Draw  if the statement is true and  if the statement is false.

1. A renewable natural resource can exist in limited amounts.

 **(Smiley / True)** — True. Although freshwater is a renewable resource, the actual amount of clean, accessible freshwater available on Earth at any single moment is limited.

2. A renewable natural resource can never become scarce.

 **(Sad Face / False)** — False. If a renewable resource (such as water or forest) is consumed or polluted faster than natural processes can replenish it, it can easily become scarce.

3. Water, if exhausted, cannot be renewed.

 **(Sad Face / False)** — False. Water is continually circulated and renewed through the natural water cycle (evaporation, condensation, rain, and snowfall).

4. Water occurs in unlimited amounts because it is renewed through rain.

 **(Sad Face / False)** — False. Although rain renews water, it does not provide an unlimited supply of instant, usable water. Overuse and lack of rain lead to drought and dry lakes.

5. Rivers and lakes will never dry.

 **(Sad Face / False)** — False. Rivers, wells, and lakes can completely dry up if there is prolonged low rainfall or severe over-extraction for human, industrial, and agricultural use.

6. We must use water and other renewable resources carefully.

 **(Smiley / True)** — True. Careful, conservative consumption ensures that natural resources have sufficient time to be replenished and do not suffer temporary or permanent exhaustion.